

Home assignment: day 11 → day 14

Deadline: 2026-07-16, 16:59. You can use sympy or any other software for symbolic computations.

1. Let's consider a 3×3 mosaic of no more than k colors. We consider two mosaics the same if one can be obtained from the other using rotations or flips.
 - a) Explicitly describe the group of actions G acting on mosaics.
 - b) How many different mosaics are there?
 - c) Describe all possible mosaics with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

2. Watch the Mirek Olšák video «Burnside's lemma: counting up to symmetries» [?].

https://www.youtube.com/watch?v=D0d9bYZ_qDY,



3. Watch the Thom de Boer video «Burnside's lemma: An Introduction to Group Theory» [?].

<https://www.youtube.com/watch?v=Exot1TqjE1Y>,



Demo for the quiz on day 12

Let's consider a 4×4 mosaic of gems where each gem can be one of k types. Consider a 90° clockwise rotation of the mosaic.

1. Represent this rotation as a sequence of cycles and write the corresponding cycle monomial.
2. How many 4×4 mosaics are fixed by this rotation?